

Grand Canyon Lidar DTM, DSM and CHM Analysis

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Abstract

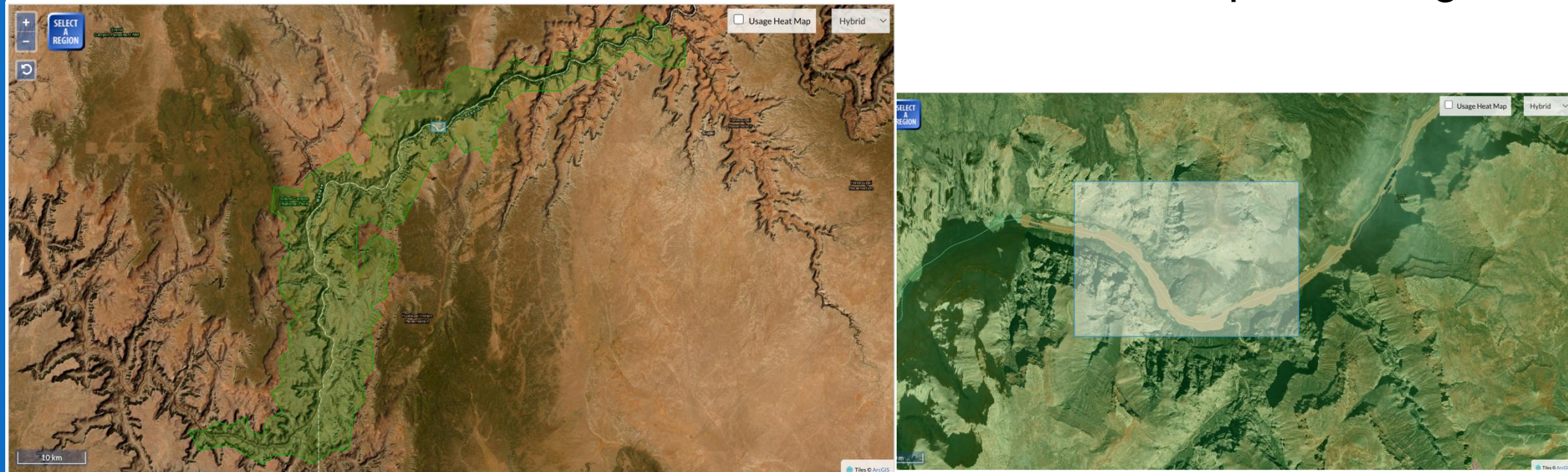
For my project I produced a Digital Terrain Model (DTM) and Digital Surface Model (DSM) and identified vegetation for a Canopy Height Model (CHM) for a riverbend region of the Grand Canyon based on lidar data collected by USGS in 2019 and published in 2022.

Introduction

The Grand Canyon is an important park as a national treasure to the country with immense geological history of how it was formed through hundreds of thousands of years of weathering and terrain changes to form the canyons that thread through the landscape. I think this could be useful for analyses to provide context for the terrain and land features for a section of the region.

Study Area and Data

This study area consisted of 31,845,366-million-points from an overall 8,894,612,025-billion-point lidar flight data surveyed from September to October 2019 and published on September 12, 2022, with a survey area of 979km ². The region of interest I selected is specifically a section of the Colorado River that runs through the Grand Canyon and includes extensive elevation change with canyon sides and rocky outcrops with a min elevation of 493.64 meters and a max elevation of 1157.5 meters. I chose this region because of the varied topographical features and to reduce the size of the LAS file for easier processing.



This data was obtained from a USGS 3DEP lidar dataset, AZ_GrandCanyonNP_2019_B19, via OpenTopography. The region of interest has a point spacing of 0.293 meters and a CRS of WGS 1984 UTM Zone 12N.

Methods

I selected the region of interest, downloaded the data in Laz format, converted it to Las with LasTools then added it to a LAS dataset in ArcGIS.

To classify vegetation, I used Classify LAS by Height on the pre classified ground and unclassified points, with parameters: unclassified (1) : 0.1m, low vegetation: 0.1 to 2m, medium vegetation: 2 to 5m and high vegetation: 5m and up.

For the DTM (Digital Terrain Model), I used binning, minimum and linear for the parameters, along with selecting ground point filters.

I used a cell size of 1.172 m, 4 times the 0.293 m point spacing.

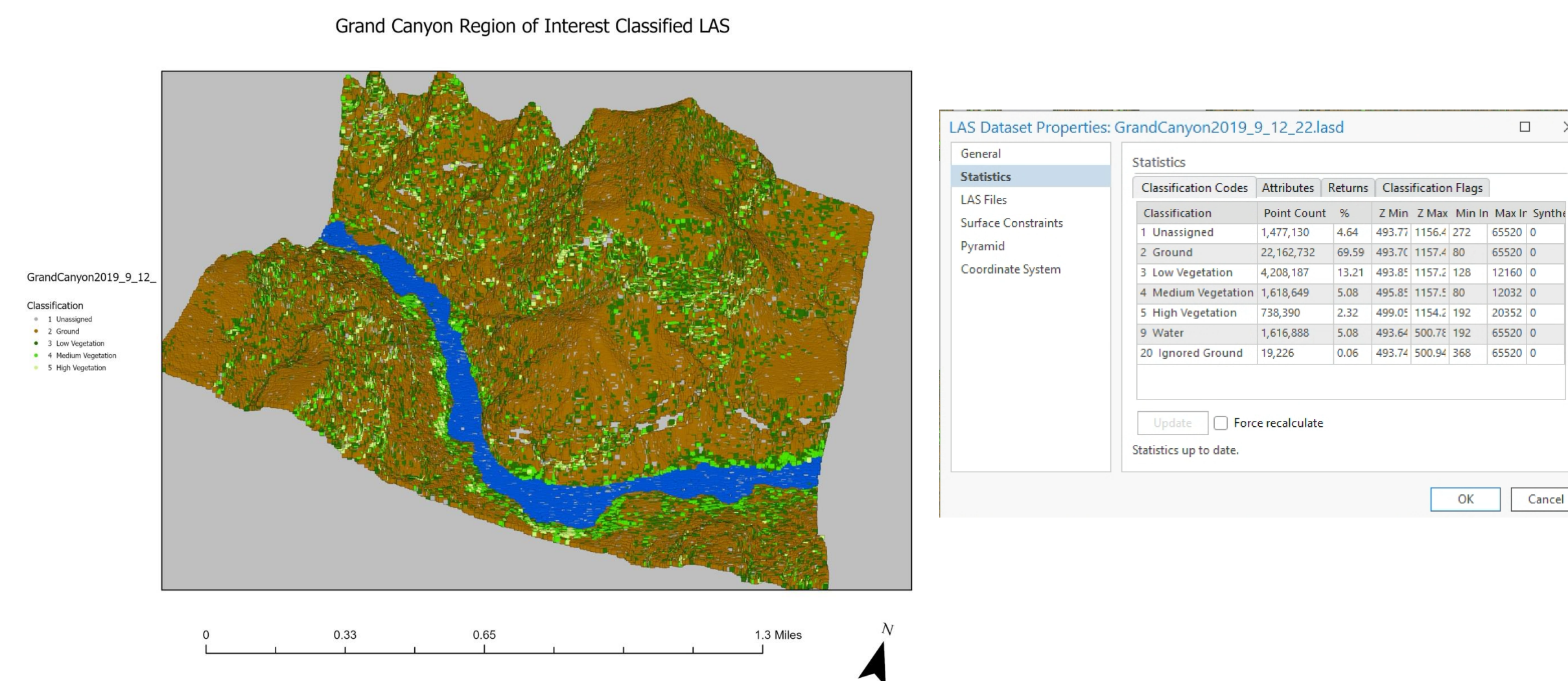
For the DSM (Digital Surface Model), I started with the classified LAS dataset filtered for 1st returns.

Then I used LAS Dataset to Raster, with parameters for value field as elevation, binning, cell assignment max, and linear, and used the same cell size as previously.

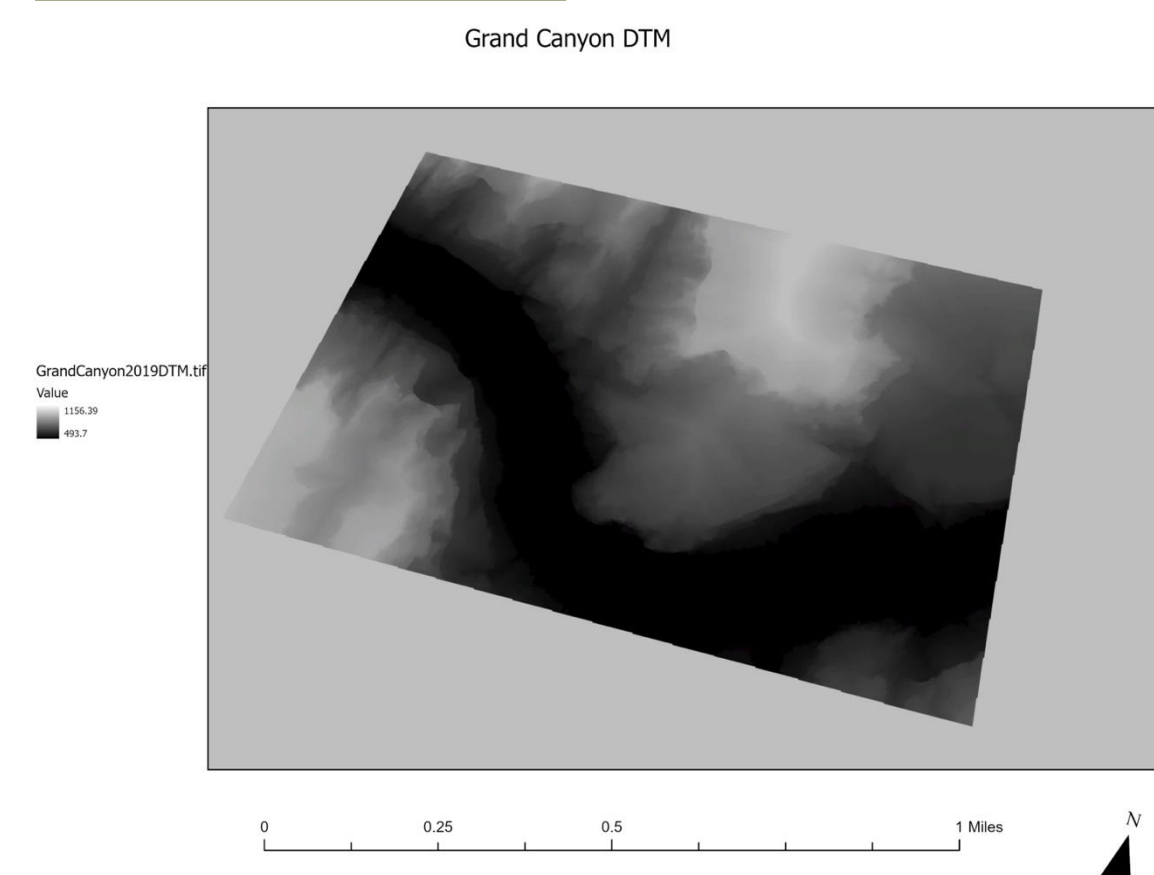
To calculate the CHM, I used raster calculator for DSM-DTM.

Finally for metrics, I converted the 3 classes of vegetation points into LAS to Multipoint and did Zonal Statistics with the previously made DTM raster to obtain statistics for the entire region of vegetation points.

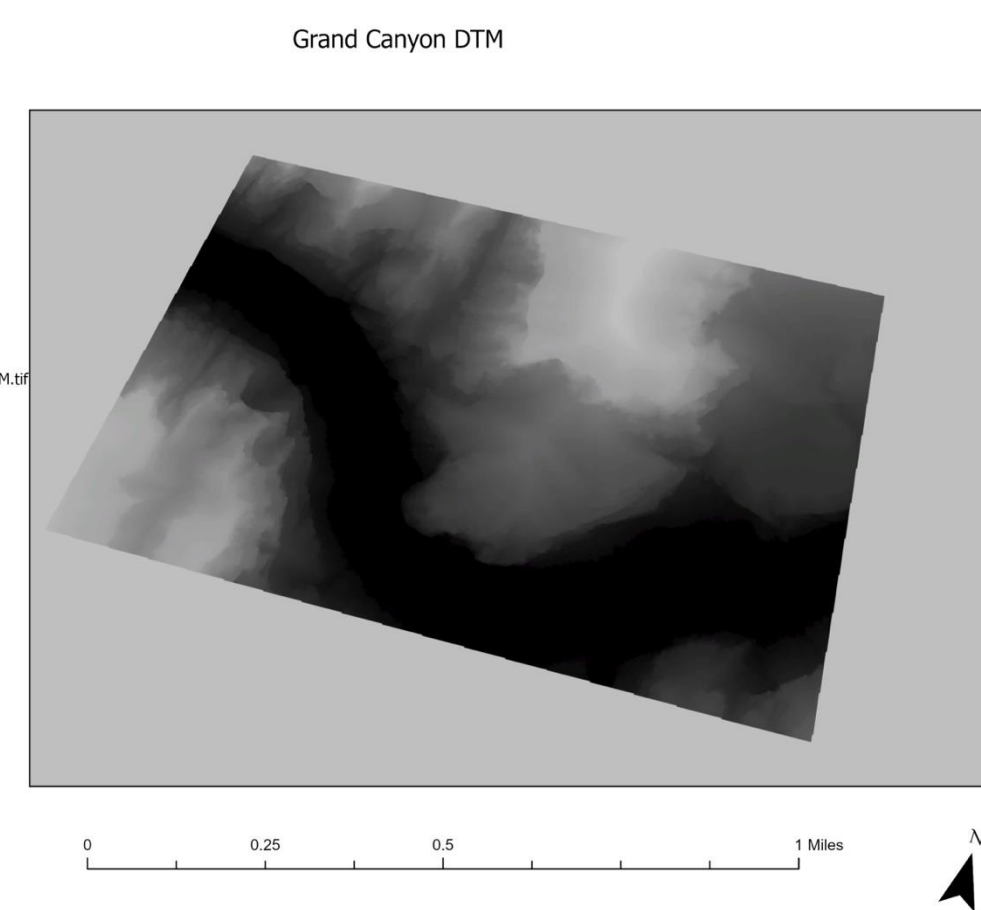
1. Vegetation Classification



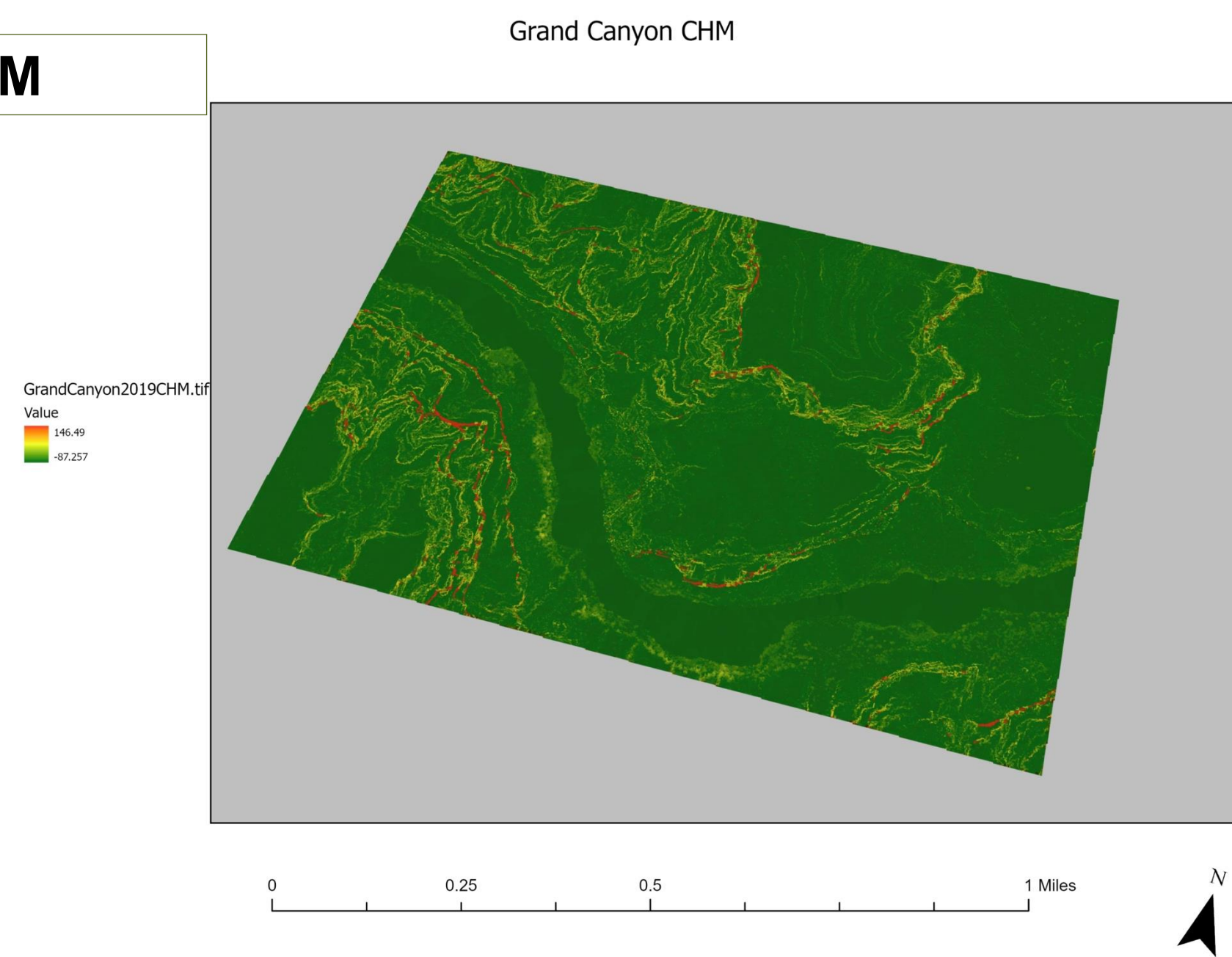
2. DTM



3. DSM



4. CHM

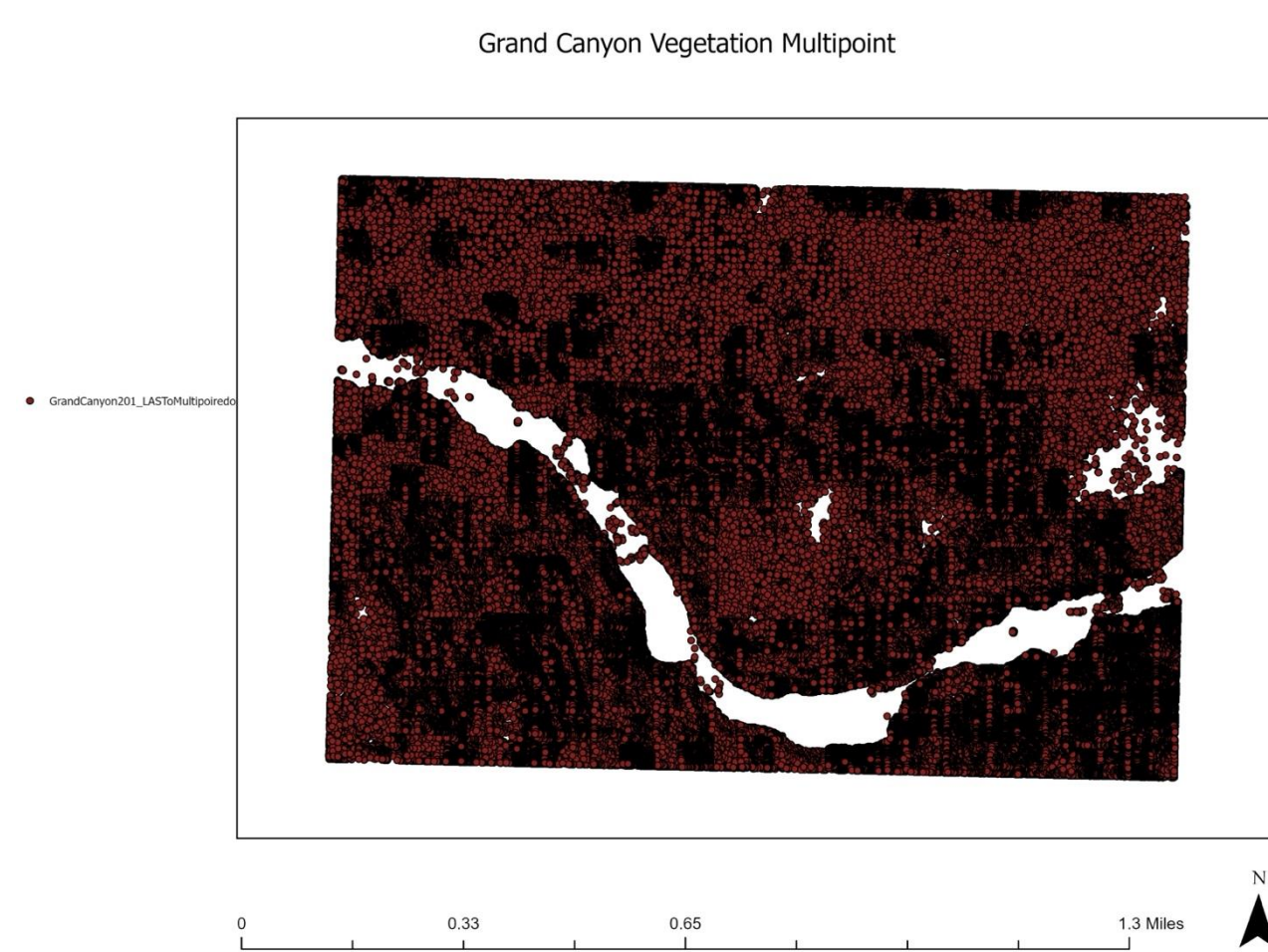


5. Zonal Statistics based on Vegetation Multipoint and DTM

GrandCanyon201_LASToMultipointredo

zonalst_gcvegetationdtm

Field:		Selection:					Rows:					
Rowid	ID	COUNT	AREA	MIN	MAX	RANGE	MEAN	STD	SUM	MEDIAN	PCT90	
1	1	0	868055	1192346.45912	493.709991	1155.02002	661.310028	737.61586	174.274939	640291135.022887	728.22998	994.66998



Results and Discussion

For the result of my analysis I found that the vegetation canopy height ranges from -87.257m in regions lacking vegetation such as the river to 146.49m at max.

In addition, I found that the entire vegetation coverage of the region of interest covers an area of 1192346.459 m².

Through Zonal Statistics metrics I found that the average point elevation for the classified LAS vegetation points was 737.616 meters and the standard deviation was about 174.275 meters.

In addition, through classification methods I found that there were 4,208,187 low, 1,618,649 medium, and 738,390 high vegetation classified lidar points, ranging from 0.1 to 2m, 2 to 5m and 5m and higher, respectively.

One thing I noticed is that handling huge lidar datasets with millions of points can be significantly impeded by processing capabilities of a computer system. One way to address this in the future could be utilizing smaller region of interests or upgrading to higher computer specifications to increase processing speeds.

Conclusion

Overall, I reasoned that areas with taller vegetation above 5 meters would likely indicate taller trees, regions below 5m but above 2m would represent middle growth trees, while regions with sub 2 meter canopy heights would likely indicate shrubbery, or low lying grasses and bushes.

In conclusion, CHM analysis can be useful to study vegetation growth on various topographical land regions such as the Grand Canyon region analyzed here. Furthermore, CHM analysis can provide insights into vegetation health, forest and ecosystem structure. More information could be gained by looking into comparing year to year results to identify canopy height changes in future research.

Sources

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